Extract Age Uncertainty from VegVault

Description

Retrieves per-sample age-model iteration data from a pre-built vaultkeepr query plan and joins the result to a caller-supplied sample-to-dataset mapping.

Usage

```
extract_age_uncertainty_from_vegvault(
  plan,
  data_sample_mapping,
  verbose = TRUE
)
```

Arguments

`plan`	A vaultkeepr plan object scoped to fossil pollen archive datasets, typically created by 'build_vegvault_plan()' with 'sel_dataset_type = "fossil_pollen_archive"'.
`data_sample_mapping`	A data frame with at least the columns 'dataset_name' and 'sample_name', used to attach dataset identifiers to the per-iteration age estimates (e.g. 'data_community_proportions').
`verbose`	Logical. If 'TRUE' (default), progress messages are printed to the console via 'cli'.

Details

The function performs the following steps: 1. Validates inputs. 2. Retrieves age uncertainty via 'vaultkeepr::get_age_uncertainty()'. 3. Pivots the wide output (one column per iteration) to long format and filters NA age estimates. 4. Joins with 'data_sample_mapping' to attach 'dataset_name'.

Input validation and plan construction (geographic/temporal filters, dataset type selection) are handled by 'build_vegvault_plan()'. Only 'fossil_pollen_archive' datasets carry per-iteration age estimates; modern plot/gridpoint datasets have no age uncertainty.

Value

A tibble with columns 'dataset_name' (character), 'sample_name' (character), 'iteration' (integer), and 'age_uncertainty' (numeric). Each row is one age-model iteration estimate for one sample. Returns an empty tibble with those columns when no age uncertainty data are available.

See Also

[build_vegvault_plan()], [extract_data_from_vegvault()],
[interpolate_paleo_community_with_age_uncertainty()]